

Shivam Kumar Singh

Staff Software Engineer | Application Architecture, Industrial SaaS, Identity, Licensing and Applied AI | .NET and React

Bengaluru, India | Remote

Phone: +91 8840330669 | Email: singhshivam071093@gmail.com

Professional Summary

Staff Software Engineer and hands-on application architect with nearly 10 years of experience building and modernising enterprise products. Progressed from full-stack implementation to technical direction and platform ownership across industrial SaaS, centralised identity, and software licensing, with hands-on contributions to governed AI systems. Combines C#/.NET and React delivery with architecture decisions spanning API boundaries, data persistence, runtime caching, authentication, authorisation, integration, containerised deployment, and operational support. Promoted to Staff Engineer at Gleason after taking cross-product responsibility and now leads five engineers while remaining hands-on with design, implementation, reviews, and complex troubleshooting. Effective at turning ambiguous product and engineering-domain requirements into reusable platforms and maintainable delivery plans across developers, QA, product owners, and domain specialists.

Core Competencies

- Application architecture and platform engineering
- Industrial SaaS and desktop-to-browser modernisation
- Enterprise identity, Single Sign-On, and application security
- Software licensing, entitlement, and runtime-state platforms
- Full-stack engineering with C#/.NET and React
- REST API contracts and enterprise integration
- Relational data, distributed caching, and containerised delivery
- Governed agentic AI and human-in-the-loop automation
- Technical direction, mentoring, and cross-functional delivery

Professional Experience

Gleason Corporation

Staff Engineer | April 2026 – Present

Senior Software Engineer | August 2021 – March 2026

Bengaluru, India | Remote

Promoted to Staff Engineer following increasing ownership of technical direction and delivery across Gleason's engineering SaaS, licensing, identity, and AI platform initiatives.

Technical Leadership and Architecture

- Lead five software engineers across engineering, licensing, identity, and AI initiatives while remaining hands-on with architecture, implementation, code reviews, debugging, and delivery.
- Define application boundaries and integration approaches across React frontends, .NET services, relational databases, Redis, identity, industrial systems, and containerised deployments.
- Convert ambiguous product and engineering-domain requirements into technical designs, API contracts, epics, implementation plans, and independently deliverable work.
- Establish reusable cross-product capabilities through shared identity, authorisation, licensing, and integration services, reducing reliance on application-specific implementations.
- Guide engineers through design trade-offs and complex defects, and coordinate technical-risk assessment, release readiness, and production troubleshooting with product, QA, and domain specialists.

KISSsoft SaaS Platform

KISSsoft supports the standards-based design, optimisation, and verification of machine elements.

- Modernise established desktop-oriented engineering workflows for browser and cloud delivery using React, Redux, ASP.NET Core, Entity Framework Core, and SQL Server.
- Define REST API contracts between the browser application and backend engineering services, creating a maintainable boundary between user workflows and domain capabilities.
- Translate complex calculation workflows with engineering-domain specialists into maintainable browser functionality and software requirements.
- Integrate the shared Gleason identity platform, replacing product-specific authentication with consistent authentication and application-scoped authorisation.

Technology: React, TypeScript, Redux, JavaScript, ASP.NET Core, Entity Framework Core, SQL Server, REST APIs, Docker

KISSsoft License Manager

- Designed and delivered a cloud-based licensing platform supporting floating and standalone KISSsoft licences, customer entitlements, usage tracking, and administrative workflows.
- Separated fast-changing runtime licence state in Redis from durable customer, entitlement, and configuration data in SQL Server, clarifying caching and persistence responsibilities.
- Implemented acquisition, validation, allocation, and release workflows used by desktop and cloud clients, with shared identity and access-control integration.
- Delivered Blazor administration capabilities, ASP.NET Core services, containerised deployment support, and cross-layer troubleshooting for licensing operations.

Technology: C#, ASP.NET Core, Blazor, Entity Framework Core, SQL Server, Redis, Docker, JavaScript

Enterprise Authentication and Single Sign-On Platform

- Engineered the reusable AuthServer and AuthServer API platform as a shared authentication authority for Gleason enterprise applications, replacing separate identity implementations.
- Enabled Single Sign-On through OpenID Connect authorisation code with PKCE for browser clients, OAuth 2.0 client credentials for service-to-service communication, and access-token validation for protected APIs.
- Consolidated users, applications, roles, claims, and application-scoped authorisation, enabling KISSsoft SaaS, KISSsoft License Manager, and other integrated products to adopt consistent security controls.
- Strengthened the platform with email OTP and TOTP multi-factor authentication, trusted-device controls, Redis-backed rate limiting, security auditing, WebAuthn/FIDO2 passkeys, and Azure Key Vault-managed secrets.

Technology: .NET 9, ASP.NET Core Identity, OpenIddict, OAuth 2.0, OpenID Connect, Entity Framework Core, SQL Server, Redis, Azure Key Vault, WebAuthn/FIDO2, Docker

GEMS AI - Governed Agentic AI Platform for Industrial Operations

- I work closely with the Principal Architect and contribute hands-on to reusable foundations for industrial data integration, AI orchestration, governed execution, and service generation.
- Implemented core agent-execution capabilities including human approval gates, resource controls, immutable audit trails, independent agent review, and resilient long-running workflows with recovery behaviour.
- Developed a service-generation pipeline supporting deterministic, interactive, and autonomous modes, creating a repeatable path from service definition to deployable output.
- Contributed to enterprise and shop-floor integration through SAP OData and resilient MQTT/Mosquitto messaging, connecting central business data with edge environments.
- Strengthened platform quality through comprehensive automated unit and integration testing, improving confidence in platform changes and generated services.

Technology: C#, .NET 8, Microsoft.Extensions.AI, MQTT, Mosquitto, PostgreSQL, pgvector, SAP OData, xUnit, NuGet

DealerSocket, a Solera Company

Software Engineer I | November 2019 – August 2021

Bengaluru, India

- Delivered automotive CRM functionality across React, TypeScript, Angular, C#, ASP.NET Web API, ASP.NET Web Forms, VB.NET, and SQL Server, spanning user workflows, APIs, business logic, and stored procedures.
- Advanced React adoption for important CRM features while maintaining integrations with established Web Forms and VB.NET modules.
- Resolved defects across frontend, backend, and data layers, including longstanding local-environment problems that impeded engineering productivity.
- Served as Scrum Master, facilitating planning, refinement, daily coordination, reviews, and retrospectives across developers, QA, and product stakeholders.
- Earned multiple **Star of the Month** awards for rapid React adoption and contributions to important CRM capabilities.

Allscripts Healthcare Solutions

Associate Software Engineer | October 2016 – November 2019

Bengaluru, India

- Delivered healthcare practice-management functionality spanning administrative, financial, reporting, and operational workflows using C#, VB.NET, VB6, T-SQL, SQL Server, and Crystal Reports.
- Modernised selected legacy capabilities for browser and cloud delivery using React, Angular, .NET Core Web APIs, Docker, and Azure SQL.
- Resolved application, database, reporting, and legacy-integration defects and supported L3 teams with critical development and customer issues.
- Extended technical influence through training on Azure, Web APIs, Entity Framework, and Angular, and mentored engineers on architecture and development practices.
- Received the **Allscripts Spot Award** and **CLEAR Award** and contributed to the Allscripts Innovation Champions Club.

Technical Skills

Primary Languages: C#, TypeScript, JavaScript, T-SQL
.NET and Backend: .NET 8, .NET 9, ASP.NET Core, ASP.NET Core Identity, OpenIddict, ASP.NET Web API, Entity Framework Core, REST APIs, Blazor
Frontend: React, TypeScript, Redux, Angular, JavaScript, HTML, CSS
Identity and Security: OAuth 2.0, OpenID Connect, PKCE, JWT, Single Sign-On, role- and claims-based authorisation, MFA, TOTP, WebAuthn/FIDO2, passkeys
Data: SQL Server, Azure SQL, PostgreSQL, Redis, pgvector
Cloud and Delivery: Microsoft Azure, Azure Key Vault, Azure DevOps, Azure Container Registry, Application Insights, Docker
Applied AI: Microsoft.Extensions.AI, agentic AI orchestration, governed execution, human-in-the-loop workflows
Industrial Integration: MQTT, Mosquitto, SAP OData
Testing: xUnit, unit testing, integration testing
Architecture and Leadership: Application and platform architecture, API and integration design, technical specifications, code reviews, mentoring, Agile delivery, production troubleshooting, legacy modernisation
Previous Technologies: VB.NET, VB6, ASP.NET Web Forms, Crystal Reports

Education

Bachelor of Engineering in Computer Science and Engineering

Annamalai University, Tamil Nadu | 2012 – 2016
CGPA: 7.7/10

Languages

English | Hindi